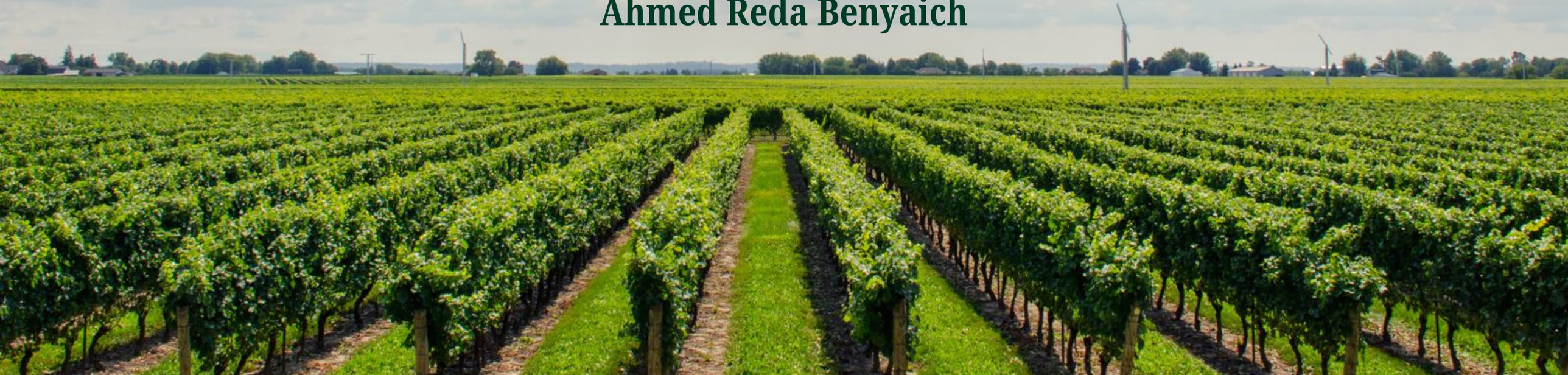
Amine Amar

CSC 4351

Predicting Crop Yield for Resilient Agri-Food Systems Using Environmental and Agricultural Data

Ahmed Reda Benyaich





Research question and objective

Research question:

Can crop yield be predicted from rainfall, fertilizer usage, land area, temperature, year, and country context?

Objective: explain and predict crop productivity

Why it matters: agriculture, planning, resilience

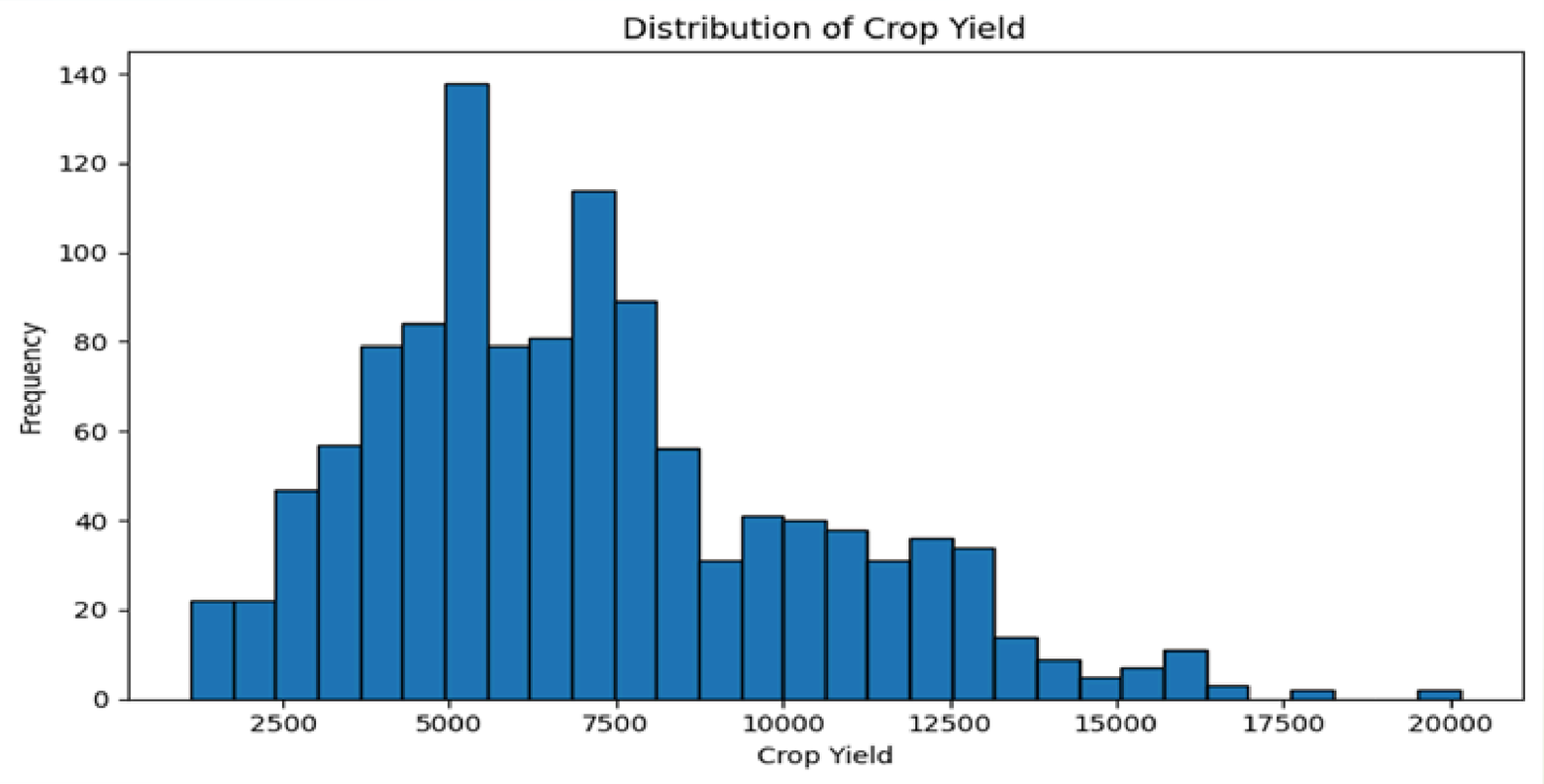
Dataset overview

- 1,172 observations
- 53 African countries
- Period: 2000–2022
- Variables: Crop_Yield, Rainfall, Fertilizer_Usage, Land_Area, Temperature, Year, Country





Milestone II: EDA main findings



Key observations

- Most values are between 4,000 and 8,000
- Distribution is right-skewed
- A few observations have very high yield

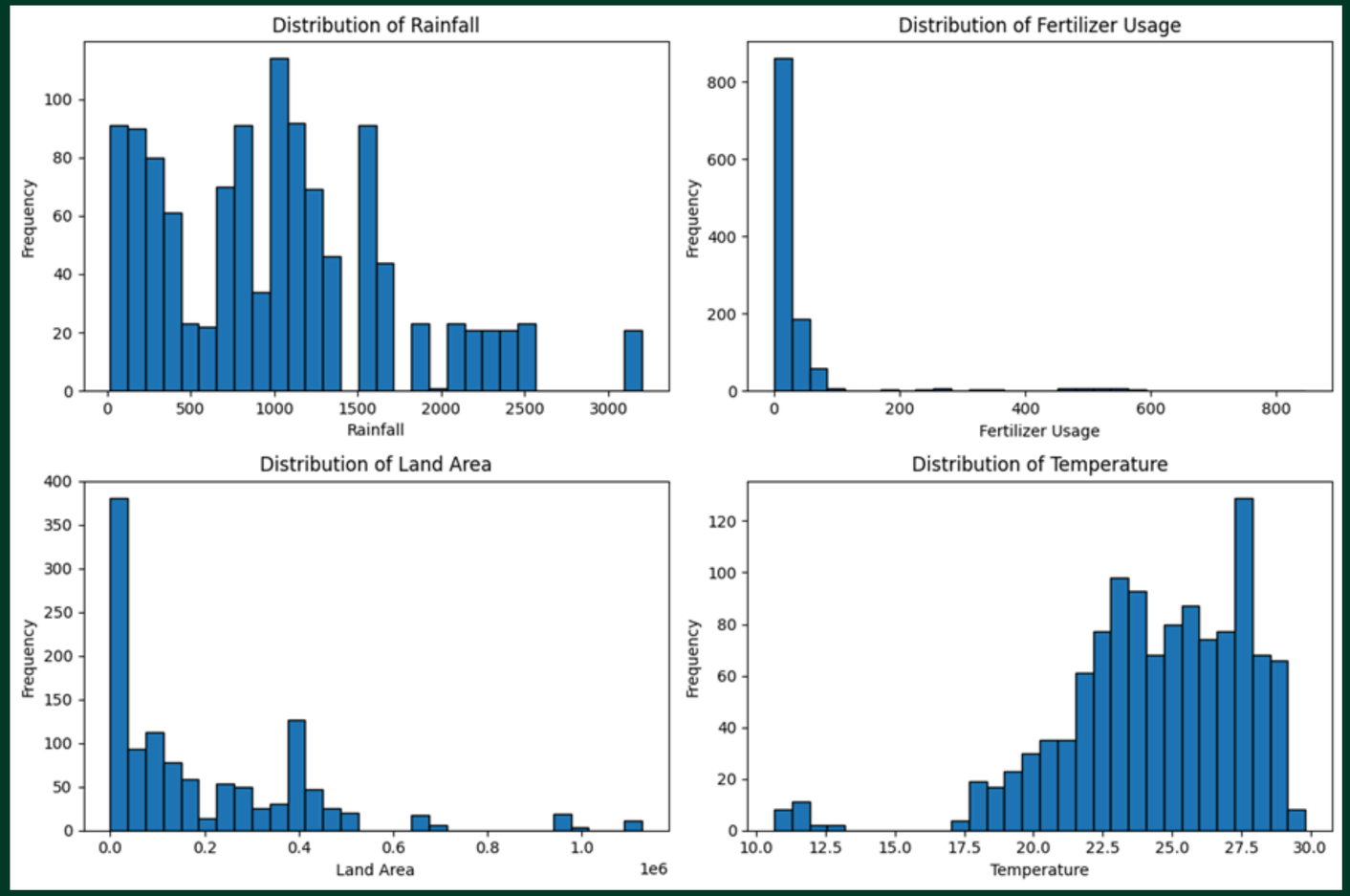
Takeaway

- Crop yield varies a lot across country-year observations

Distributions of the explanatory variables

Rainfall
- Wide spread across observations
- No very simple single pattern
- Strong geographical variability

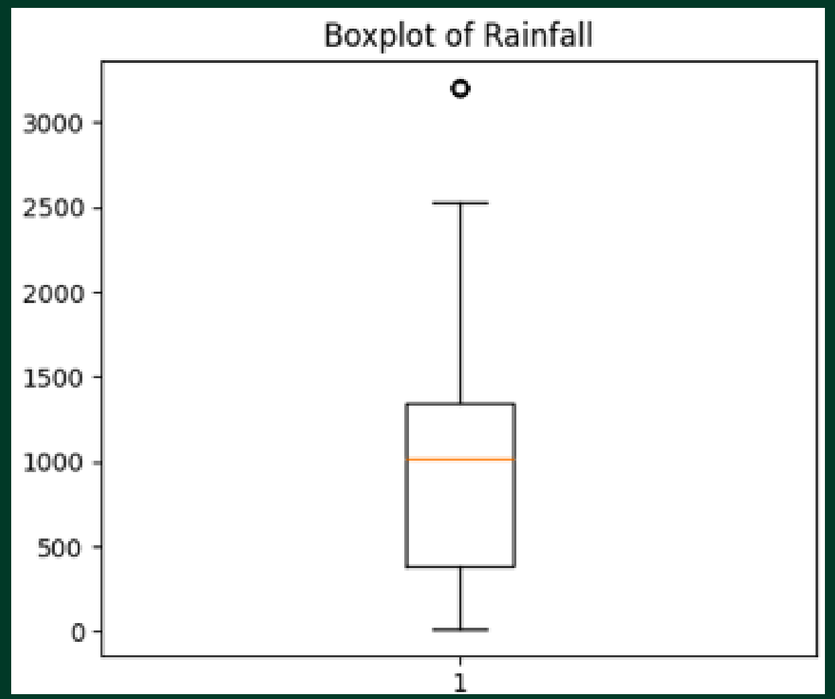
Land Area
- Also strongly right-skewed
- Large values stretch the scale
- Most observations are lower relative to the maximum



Fertilizer Usage
- Strongly right-skewed
- Most observations are at low values
- A few observations are very high

Temperature
- More concentrated
- Most values lie around 22–28°C
- Less dispersed than the other variables

Box-Plots

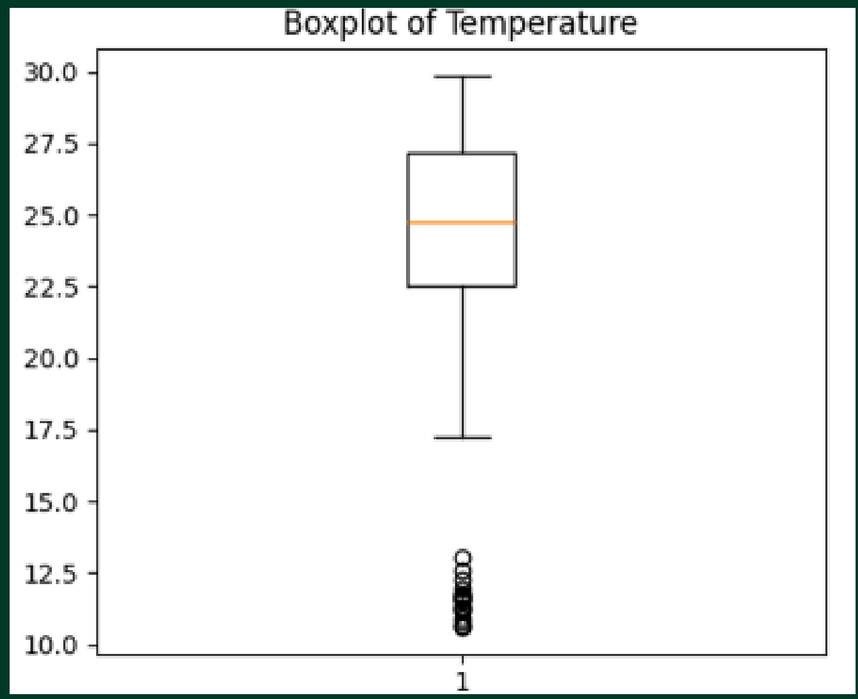


Rainfall

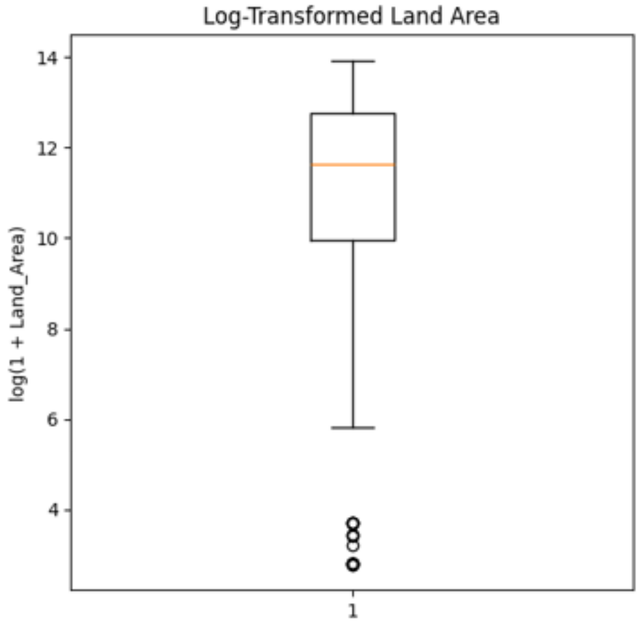
- Wide spread across observations
- A few high outliers
- Strong variability across country-years

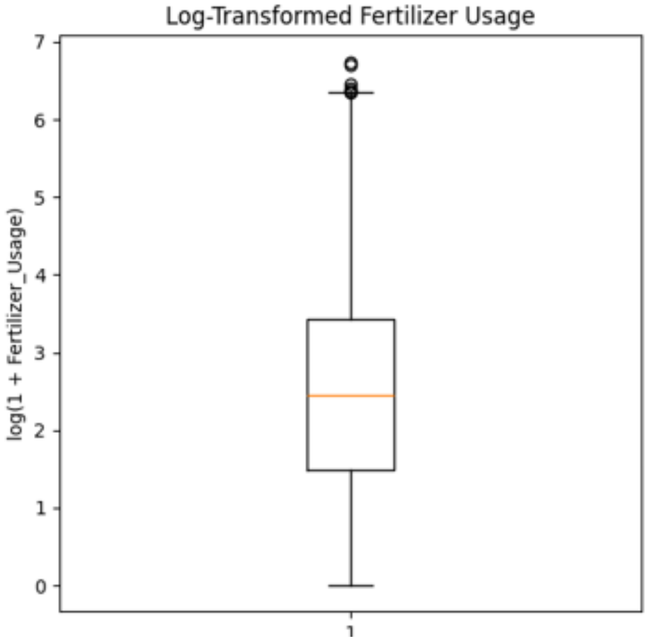
Temperature

- More concentrated than other variables
- A few low outliers
- Relatively stable distribution



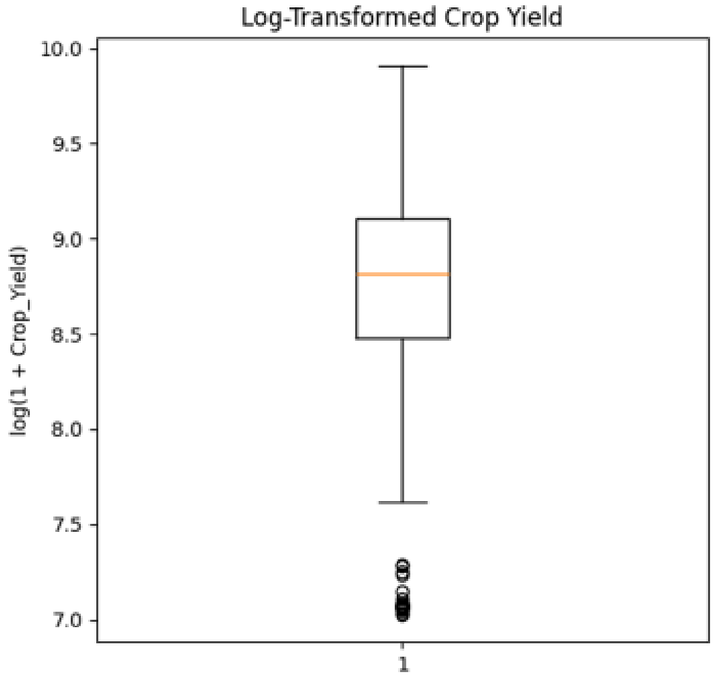
Log-Transformed Boxplots





Log boxplots (Crop_Yield, Fertilizer_Usage, Land_Area)

- Log scale reduces extreme-value distortion
- Makes the central distribution easier to read
- Confirms strong right-skewness in raw data



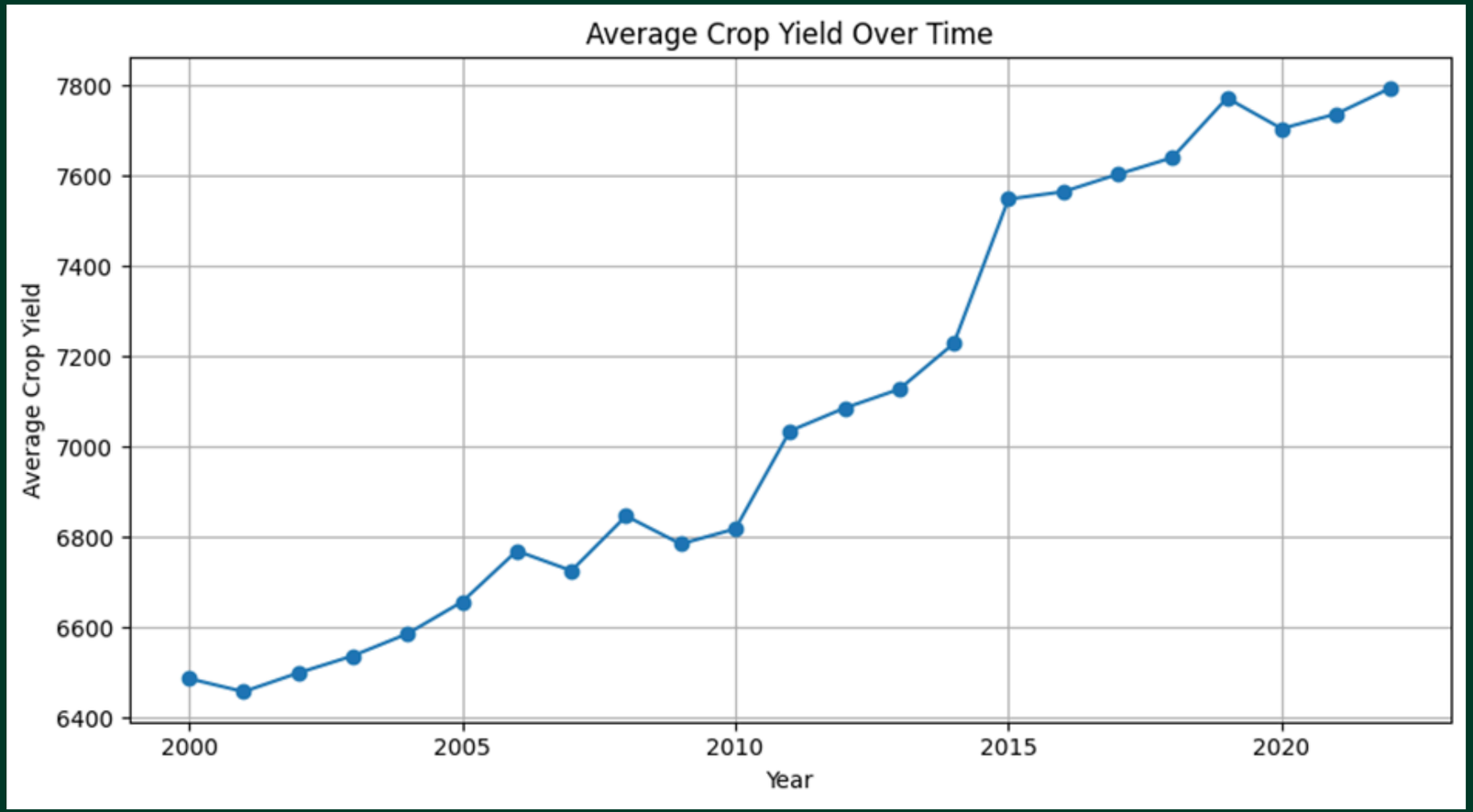
Temporal Evolution of Average Crop Yield (2000–2022)

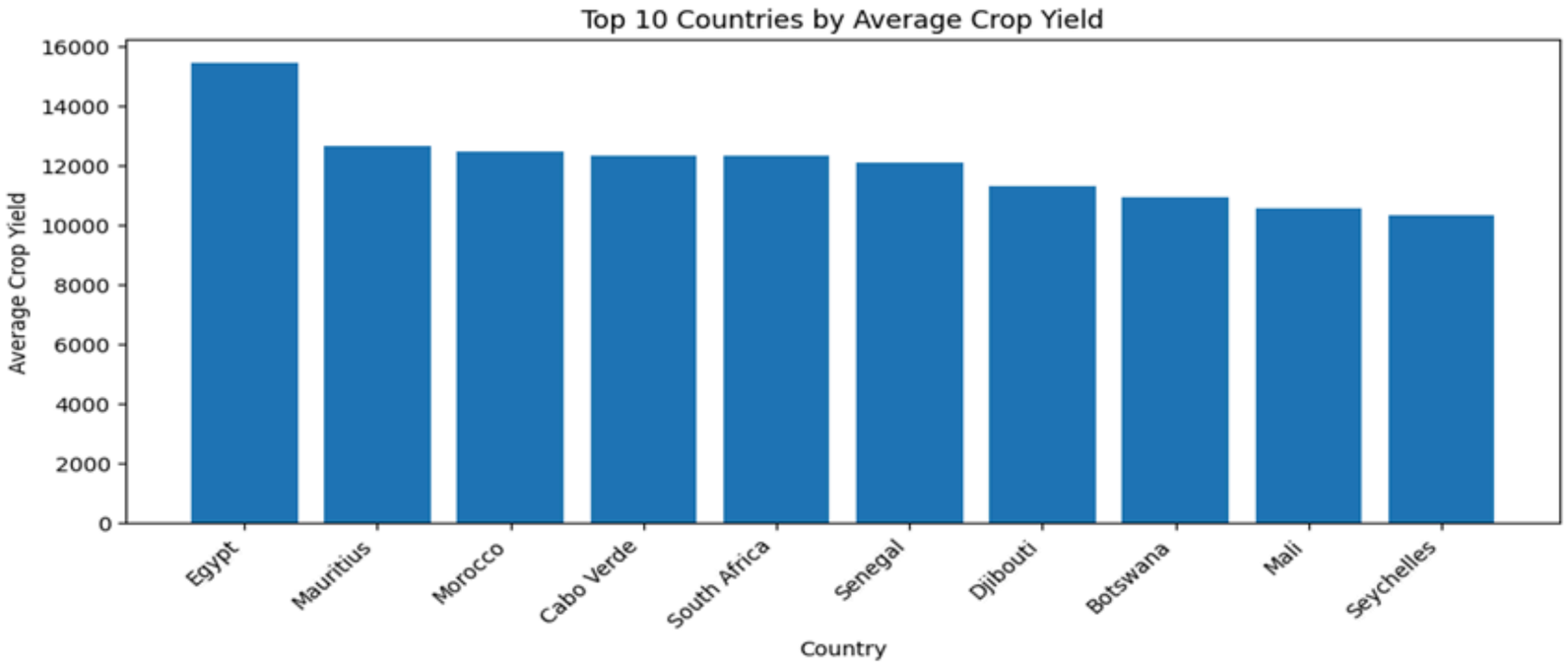
Key observations

- Clear upward trend from 2000 to 2022
- Minor fluctuations in a few years
- Stronger increase after around 2010

Takeaway

- Crop yield generally improves over time
- Year may be an important explanatory variable





Top 10 Countries by Average Crop Yield

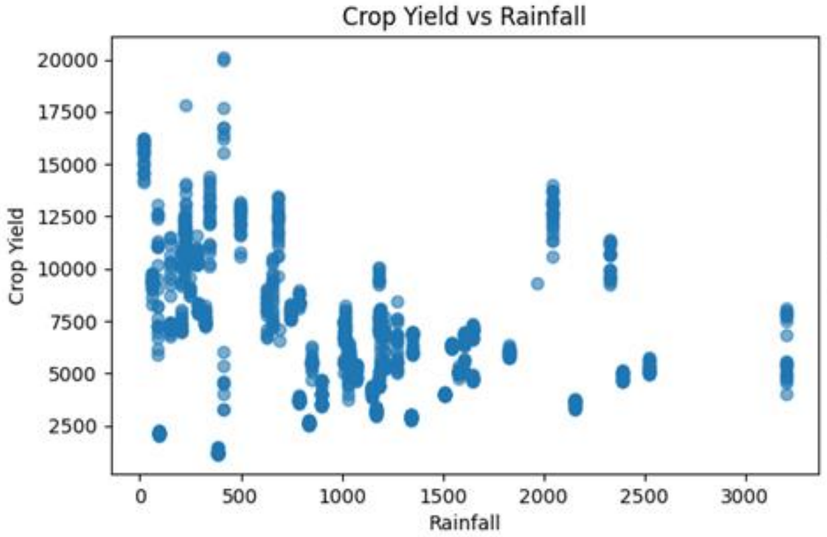
Key observations

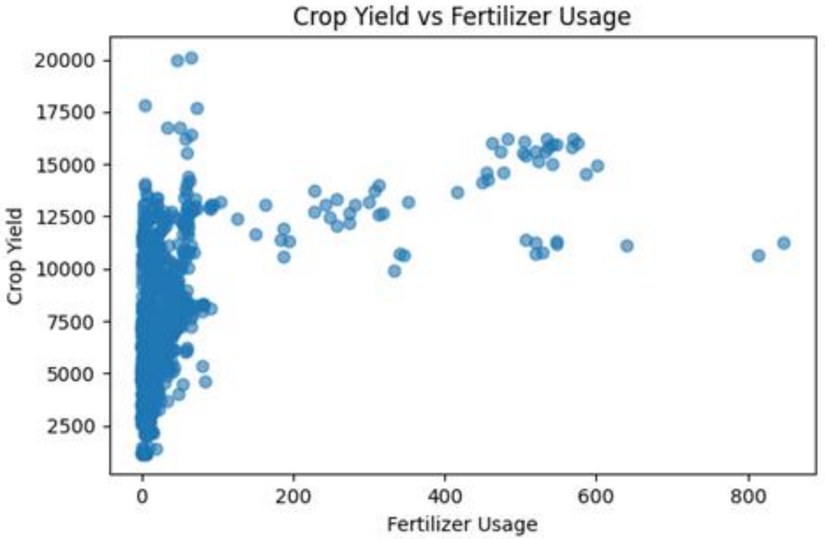
- Egypt has the highest average crop yield
- The top 10 countries show some differences in yield levels
- High-yield countries are spread across different African regions

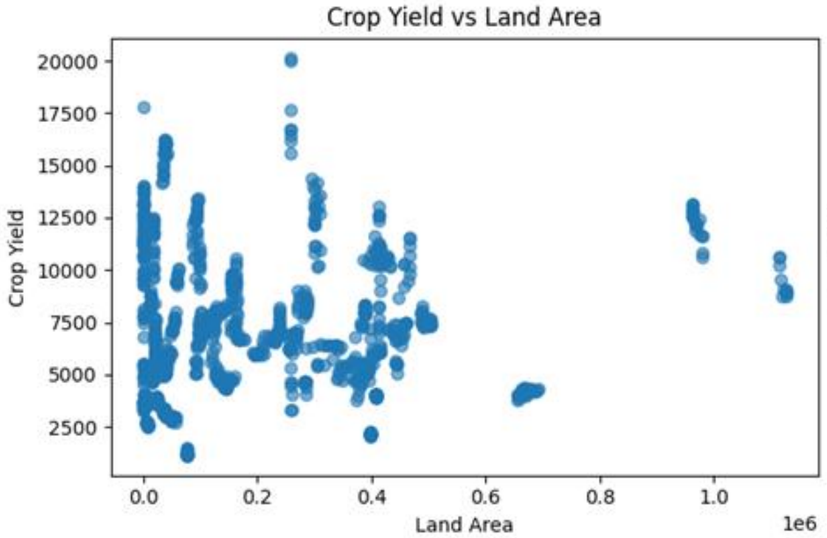
Takeaway

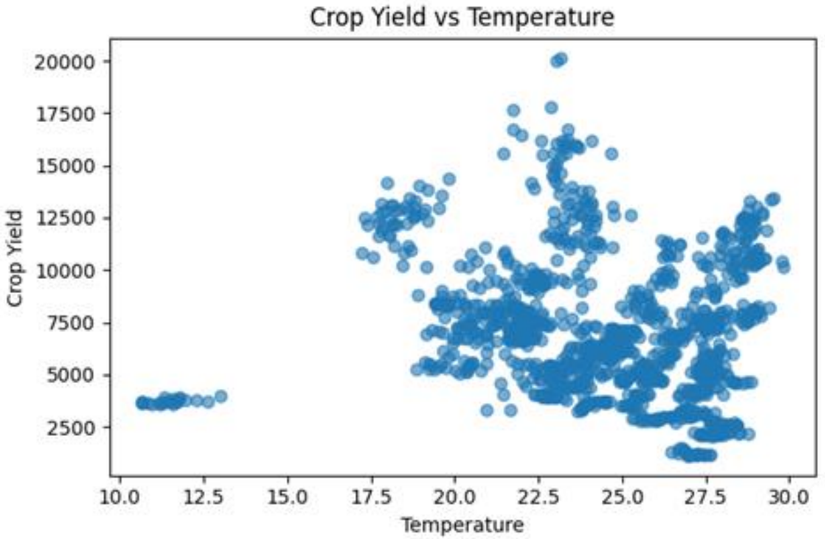
- Crop yield varies strongly across countries
- Country context is an important source of variation

Scatter Plots of Crop Yield vs Explanatory Variables









Rainfall

- Wide dispersion
- No strong clear linear trend
- Possible country-specific effect

Fertilizer Usage

- Clearest positive association
- Higher usage often linked to higher yield
- Still substantial dispersion

Land Area

- No strong direct linear pattern
- Size alone does not explain productivity

Temperature

- Clustered / complex pattern
- Relationship appears non-linear

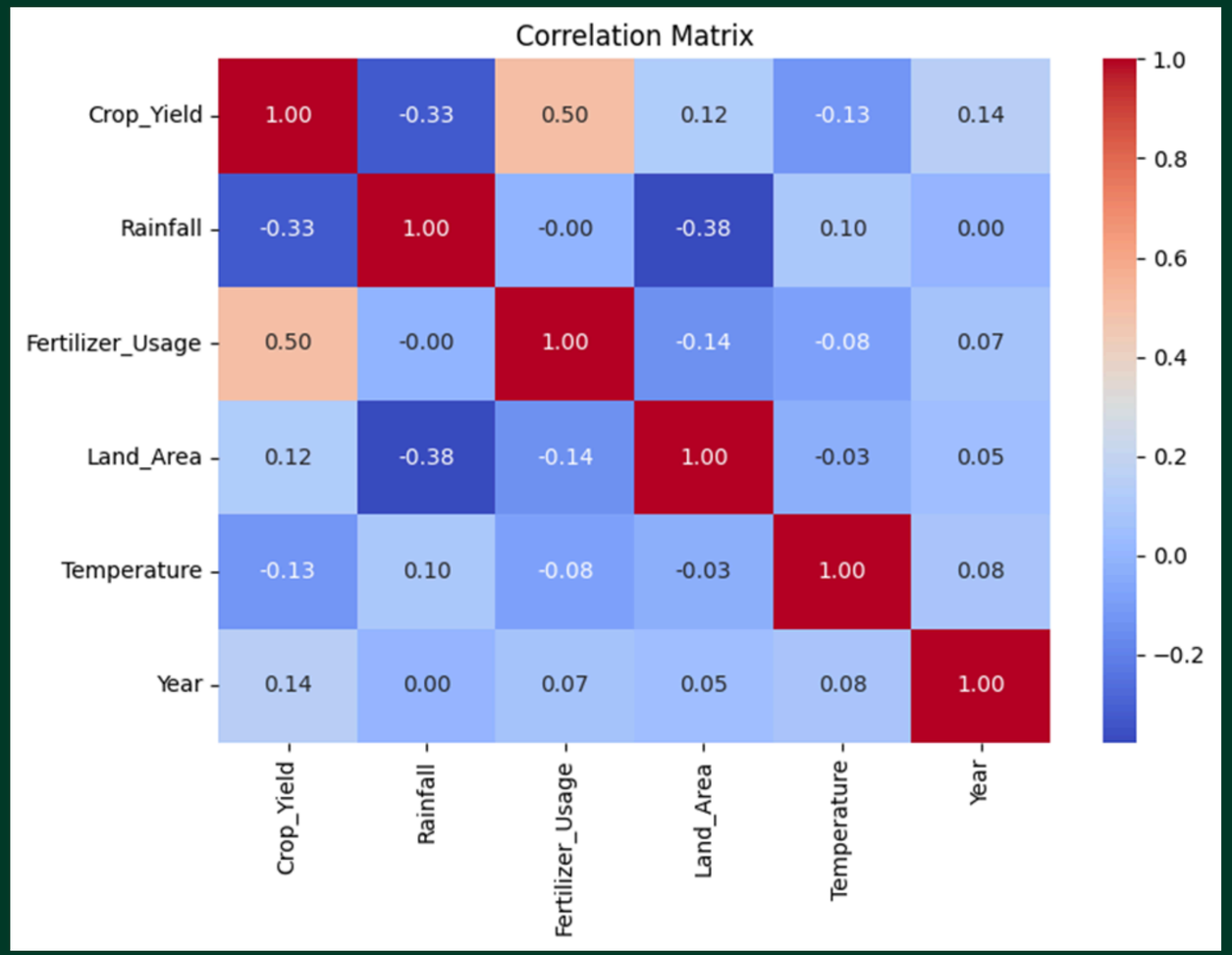
Correlation Matrix of Crop Yield and the Explanatory Variables

Key observations

- Fertilizer_Usage has the strongest positive correlation with Crop_Yield
- Rainfall shows a moderate negative correlation
- Land_Area, Temperature, and Year have weaker correlations

Takeaway

- Fertilizer_Usage is the most informative predictor in simple linear terms
- The other variables may still matter through joint or non-linear effects



Key EDA interpretations for modeling

Main findings

- Crop_Yield, Fertilizer_Usage, and Land_Area are right-skewed
- Fertilizer_Usage and Land_Area contain strong outliers
- Yield differs substantially across countries
- Relationships are not purely linear

Modeling implications

- Apply log transformation to highly skewed variables
- Keep Year and Country as contextual predictors
- Use MLR as a baseline, but interpret results carefully

Takeaway:

EDA showed that the data is heterogeneous, skewed, and partly non-linear, so preprocessing and contextual variables are important for modeling.



Variable Declaration

Response and Explanatory Variables:

- Y: Crop_Yield
- X: Rainfall, Fertilizer_Usage, Land_Area, Temperature
- Contextual variables: Year, Country

- Crop_Yield is the target variable
- The predictors describe environmental and agricultural conditions
- Year and Country capture time and country heterogeneity





Preliminary Model: Multiple Linear Regression

Why MLR?

- Crop_Yield is a continuous variable
- Coefficients are easy to interpret
- Suitable for both prediction and inference
- Can include Year and Country

Main assumptions

- Linearity
- Independence of observations
- Homoscedasticity
- Normality of residuals
- Low multicollinearity

Data Preparation for Modeling

- No missing values in selected variables
- Log transformation: Fertilizer_Usage and Land_Area
- Country encoded as dummy variables
- 80/20 train-test split
- Training set: 937 observations
- Test set: 235 observations
- Same preprocessing used for all models

Bottom takeaway:
Preprocessing choices were directly based on the EDA findings.



Model Comparison Using Cross-Validation

Model	CV R^2	CV MAE	CV RMSE
CatBoost	**0.937**	**503.7**	**806.18**
XGBoost	0.934	472.47	814.94
LightGBM	0.931	436.07	820.78
Random Forest	0.921	349.66	859.22
MLR	0.9	598.79	1022.02

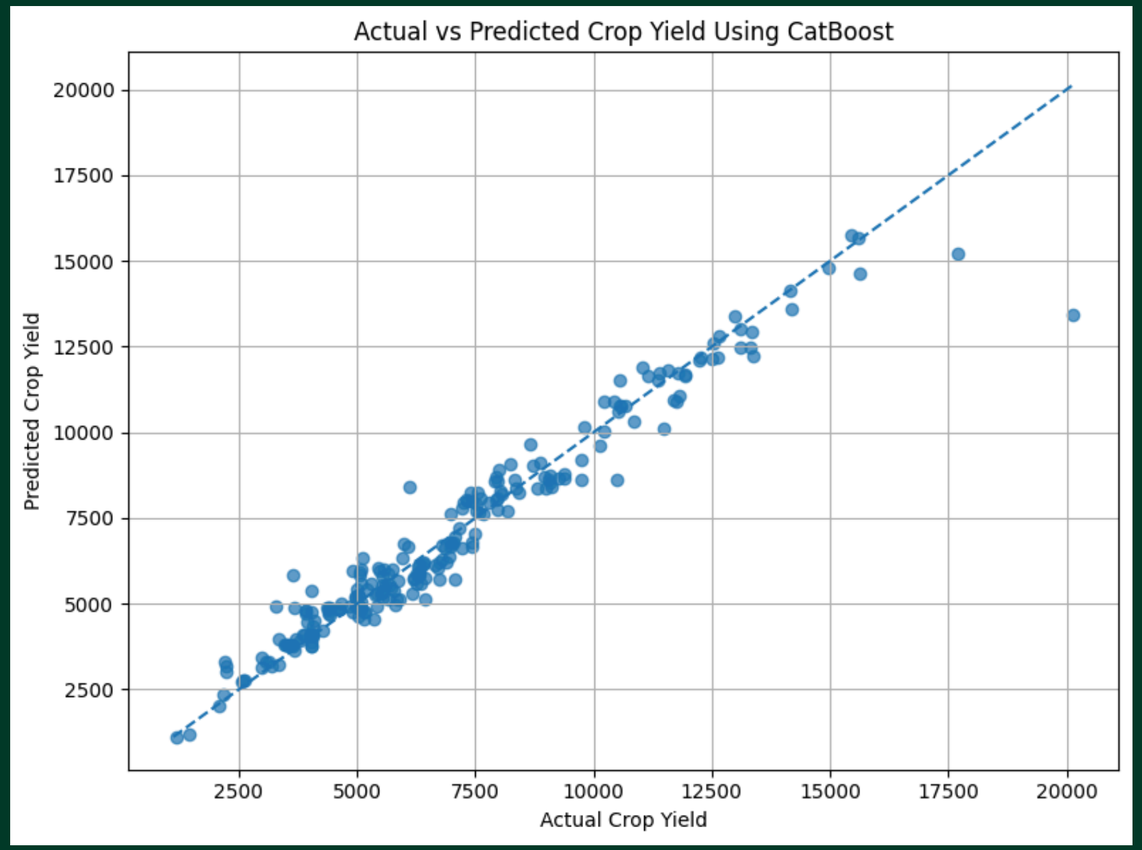
Bottom takeaway:
CatBoost was selected because it provides the best overall predictive performance.

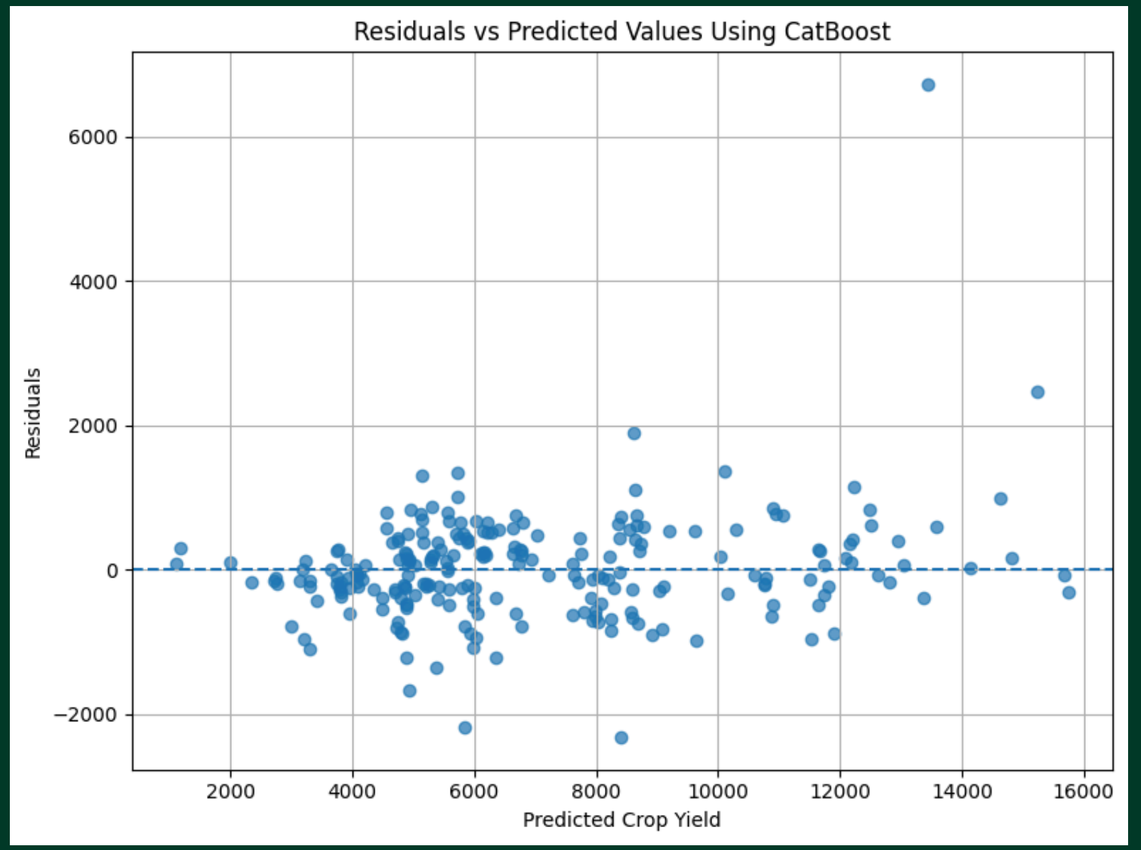


Why CatBoost?

- Captures non-linear relationships
- Handles country-level patterns
- Better aligned with the prediction objective

Prediction quality





Bottom takeaway:

Most predictions are close to the ideal prediction line, showing strong predictive accuracy.

Limitations

- Local/regional differences inside countries are not captured
- Important variables are missing: soil quality, irrigation, crop type, pests, drought events
- CatBoost is strong for prediction, but it is less interpretable than MLR
- The model is predictive, not causal



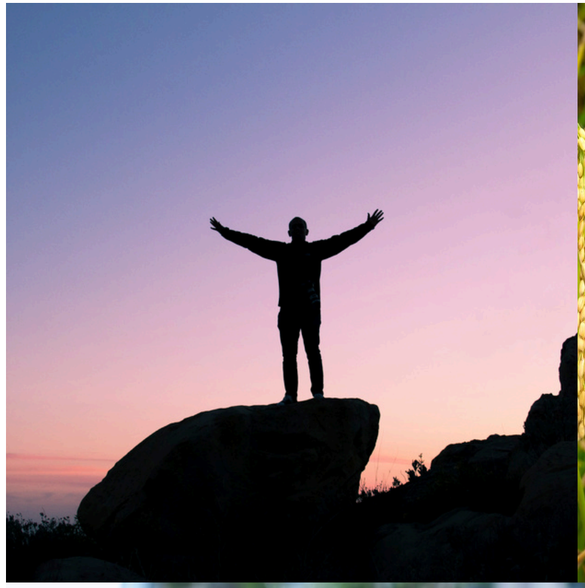


Future Improvements

- Add more agricultural variables: soil quality, irrigation, crop type, seed quality
- Add climate shock variables: droughts, floods, extreme temperatures
- Use regional or subnational data instead of only country-level averages

Conclusion

- Crop yield can be predicted well from the selected variables
- MLR was useful as an interpretable baseline
- CatBoost achieved the best overall predictive performance
- The prediction graphs show that most predicted values are close to the actual crop yield values
- The residual plot does not show a strong systematic pattern, suggesting good generalization
- Future work: add soil quality, irrigation, crop type, drought events.








Thank You